Statistical Learning and Machine Learning

Exam Topics - January 2025

TOPIC 1

You are provided with a dataset x containing the following 6 points in 1D space

$$x = \{1.0, 1.5, 1.2, 5.8, 6.2, 6.5\}$$

These points are assumed to be generated from a **Gaussian Mixture Model (GMM)** with 2 clusters (components). The corresponding GMM distribution is

$$p(x) = \sum_{k=1}^{2} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$$

where μ_k is the mean of the k-th Gaussian component, σ_k^2 is the variance of the k-th Gaussian component, π_k is the mixing coefficient of the k-th Gaussian component, and $\mathcal{N}(x \mid \mu_k, \sigma_k^2)$ represents the Gaussian PDF for the k-th component:

$$\mathcal{N}(x \mid \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)$$

.

Your task is to estimate the parameters of this GMM using a single iteration of the Expectation-Maximization (EM) algorithm. To achieve that, use the following initial parameters of the GMM model to start the EM algorithm, where the subscript $^{(0)}$ indicates the parameters at the 0-th iteration:

- **Cluster 1:** - Mean: $\mu_1^{(0)} = 2.0$ - Variance: $\sigma_1^{2(0)} = 0.1$ - Mixing Coefficient: $\pi_1^{(0)} = 0.5$
- **Cluster 2:** - Mean: $\mu_2^{(0)} = 3.0$ - Variance: $\sigma_2^{2(0)} = 0.1$ - Mixing Coefficient: $\pi_2^{(0)} = 0.5$

1) (Score: 15%) Briefly write the steps (with relevant formulas) that you would perform to estimate the means $\mu_1^{(1)}$ and $\mu_2^{(1)}$, variances $\sigma_1^{2(1)}$ and $\sigma_2^{2(1)}$, and the mixing coefficients $\pi_1^{(1)}$ and $\pi_2^{(1)}$ of the GMM model during the first iteration of the EM algorithm.
2) (Score: 60%) Using the above steps, perform a **single iteration** of the EM algorithm on the given dataset, using the initial parameters above, to compute the updated model parameters (means $\mu_1^{(1)}$ and $\mu_2^{(1)}$, variances ($\sigma_1^{2(1)}$ and $\sigma_2^{2(1)}$, and mixing coefficients ($\pi_1^{(1)}$ and $\pi_2^{(1)}$). Show the relevant steps (or code) and the updated parameter values.
3) (Score: 15%) Given the input data points x and your obtained estimates in the previous step, comment on whether the updated mean estimates appear to be converging toward values that adequately represent different clusters in the dataset.
4) (Score: 10%) Explain why an *iterative approach (EM algorithm)* is required to estimate the GMM model parameters using the Maximum Likelihood (ML) approach, unlike a standard Gaussian case where model parameters are obtained *directly*.

TOPIC 2

Consider the following dataset $\mathbf{X}$ in 2D space, containing $N = 6$ *training* data points shown by blue-circles and red-crosses in Figure 1.

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 4 & 6 & 7 & 7 \\ 3 & 5 & 3 & 6 & 7 & 5 \end{bmatrix} \tag{1}$$

The blue-circled data points belong to class $\mathcal{C}_1$ (with target $t = 1$), whereas the red-crossed points belong to $\mathcal{C}_2$ (with target $t = 0$). Using this training data, you are required to classify the *test data point*, T=$\begin{bmatrix} 5 \\ 4.5 \end{bmatrix}$ shown as a green triangle in Figure 1, to either $\mathcal{C}_1$ or $\mathcal{C}_2$, by using the **probabilistic generative classification model**.

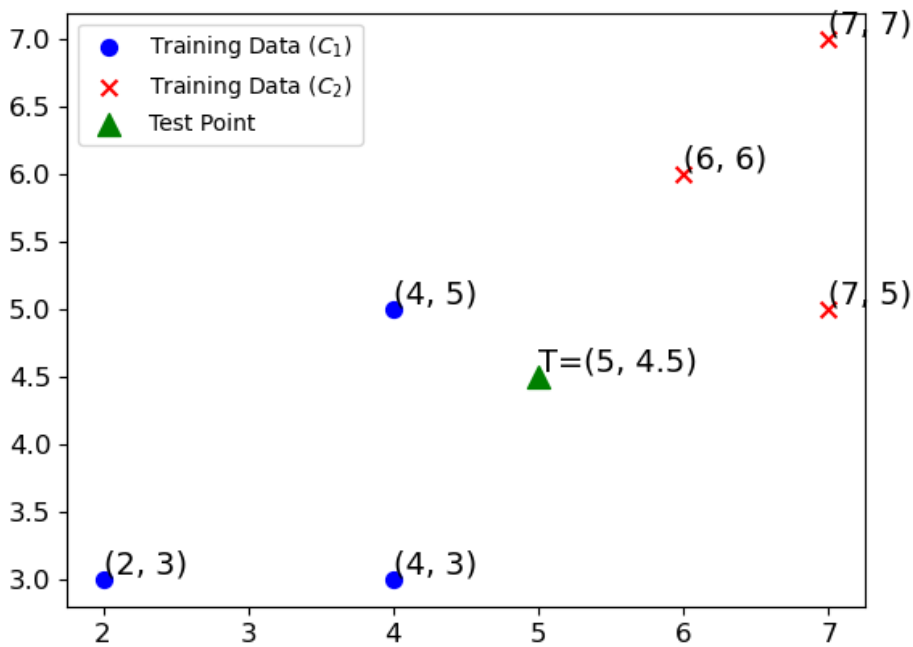

Fig. 1. The graphical illustration of the dataset (both training and test sets) used in Topic 2.

1) (Score: 50 %) Assume that the class conditional probability densities, i.e., $p(\mathbf{x}|\mathcal{C}_1)$ and $p(\mathbf{x}|\mathcal{C}_2)$ are Gaussian with means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ respectively and the *same* covariance matrix $\boldsymbol{\Sigma}$. Further, let $p(\mathcal{C}_1)$ and $p(\mathcal{C}_2)$ denote the class priors for the classes $\mathcal{C}_1$ and $\mathcal{C}_2$, respectively. Using the training data points (blue circles and red crosses in Figure 1 and the **maximum likelihood approach**, calculate the values of the parameters $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$, $\boldsymbol{\Sigma}$, $p(\mathcal{C}_1)$, and $p(\mathcal{C}_2)$ for the probabilistic generative model.
2) (Score: 25 %) Using the estimated parameter values in the previous step, compute the posterior probability function $p(\mathcal{C}_1|\mathbf{x})$.
3) (Score: 25 %) Estimate and plot the *decision boundary* of the obtained probabilistic generative classifier, together with the available data points in Figure 1. Which class will you assign to the test data point (green triangle) using the obtained decision boundary?

TOPIC 3

Consider the data points shown in Figure 2. The points are located in the 2D space and constitute training points (blue squares and red diamonds) and test points (black circles). It is known that the blue squares belong to the class C_1 and red diamonds belong to the class C_2.

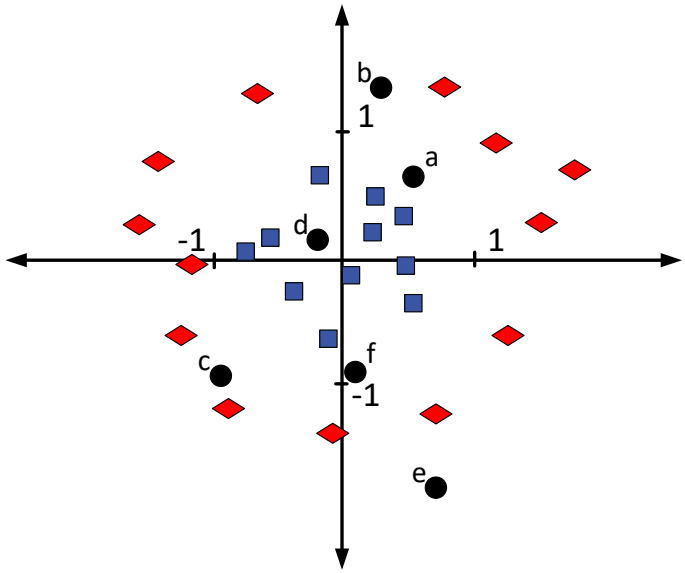

Fig. 2. Input training and test data points plotted in the 2D space. Points represented by blue squares and red diamonds belong to the classes C_1 and C_2 respectively. Six test data points are represented by black circles and symbols a-f.

1) (Score: 10%) Can you apply **Fisher's linear discriminant** model directly to input data (i.e., without using nonlinear basis functions $\phi(\boldsymbol{x})$) to *correctly* classify the test data points into the two classes, C_1 and C_2? If yes, briefly explain how that can be done. If not, why not?
2) (Score: 10%) Can you apply **Logistic regression** model directly to input data (i.e., without using nonlinear basis functions $\phi(\boldsymbol{x})$) to *correctly* classify the test data points into the two classes, C_1 and C_2? If yes, briefly explain how that can be done. If not, why not?
3) (Score: 10%) Can you apply **Linear Support Vector Machines** that uses linear kernel $\kappa(x, x') = x^T x'$ to classify the test data points into the two classes, C_1 and C_2, with *high classification accuracy*. If yes, briefly explain how that can be done. If not, why not?
4) (Score: 20%) There exists a *nonlinear transformation* $y_i = \phi(\mathbf{x}_i)$ that will allow the **Fisher's linear discriminant** and the **Logistic regression** model to work well on the above data, in the sense that both classification methods will provide 100% classification accuracy on the test data. Write down that data transformation function $y_i = \phi(\mathbf{x}_i)$.
5) (Score: 20%) After applying the nonlinear transformation $\phi(\mathbf{x}_i)$ obtained above, draw the transformed data points y_i in 1D space. Roughly sketch where the blue squares (training points from C_1 class), red diamonds (training points from C_2 class), and the black circles (test points) will reside in the transformed space. Also roughly draw the decision boundary of the Logistic regression model applied to this transformed data, and the corresponding decision boundary in the original feature space (Fig. 2). *Note that when drawing the transformed data and decision boundaries, just draw their rough arrangement in the new feature space without calculating the exact values.*
6) (Score: 20%) Which suitable kernel function could be used with **Support vector machines** to enable 100% classification of the test data points using the available training data? Identify the **Support vectors** in Fig. 2, which shows data in the original space.
7) (Score: 10%) To which class will the test data points $a, b, \ldots, f$ be classified using the SVM classifier you used to answer question 6?

TOPIC 4

Consider a single hidden-layer neural network for **binary classification task** in Fig. 3, where x_i are the nodes in the input layer, $z_{1,i}$ are the nodes of the hidden layer, and y is the single node in the output layer. $\mathbf{w}_1$ and $\mathbf{w}_2$ are the weights (including biases) connecting the input layer x_i to the hidden layer $z_{1,i}$ and the hidden layer to the output layer y, respectively. The weights of the neural network are initialized to be:

$$\mathbf{w}_1 = \begin{bmatrix} w_{0,1}^{(1)} & w_{0,2}^{(1)} & w_{0,3}^{(1)} & w_{0,4}^{(1)} \\ w_{1,1}^{(1)} & w_{1,2}^{(1)} & w_{1,3}^{(1)} & w_{1,4}^{(1)} \\ w_{2,1}^{(1)} & w_{2,2}^{(1)} & w_{2,3}^{(1)} & w_{2,4}^{(1)} \end{bmatrix} = \begin{bmatrix} 0.6 & 0.2 & -0.1 & 0.4 \\ 0.7 & -1.0 & 0.3 & 0.2 \\ -0.5 & -0.3 & -0.4 & -0.8 \end{bmatrix} \qquad \mathbf{w}_2 = \begin{bmatrix} w_{0,1}^{(2)} \\ w_{1,1}^{(2)} \\ w_{2,1}^{(2)} \\ w_{3,1}^{(2)} \\ w_{4,1}^{(2)} \end{bmatrix} = \begin{bmatrix} -0.7 \\ 0.7 \\ 0.6 \\ -0.3 \\ -0.7 \end{bmatrix}$$

where $w_{i,j}^{(l)}$ denotes the weight connecting the ith node of the $(l-1)$th layer to the jth node of the lth layer.

A logistic sigmoid activation function $z_{1,i} = h_{ls}(a_{1,i}) = \frac{1}{1+exp(-a_{1,i})}$ is used in each of the nodes of the hidden layer and in the output layer yielding the output $y = h_{ls}(a_{2,1}) = \frac{1}{1+exp(-a_{2,1})}$, where $a_{l,i}$ is the activation of the ith node of the lth layer and is given by a weighted sum of the inputs to the node. The *loss (or error) function* employed to train this network for classification is the **cross entropy error** function. The target values for this binary classification task are $t \in \{0, 1\}$.

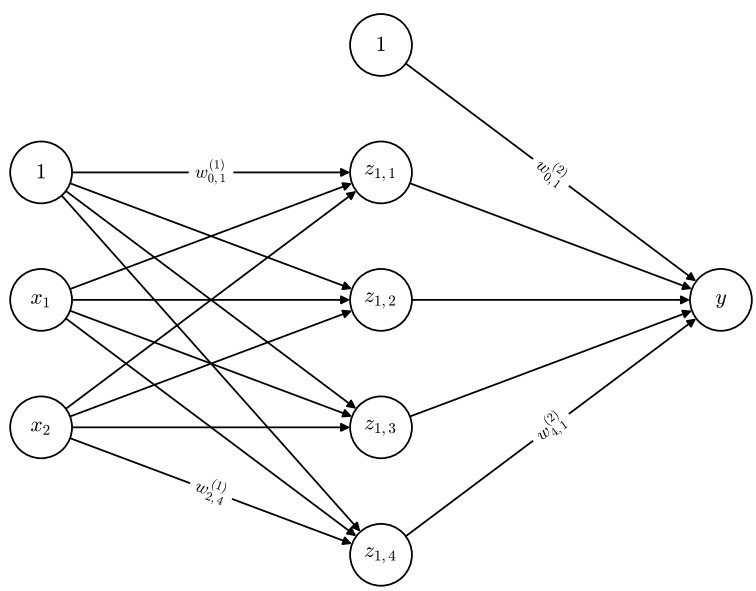

Fig. 3. A single hidden-layer neural network for classification.

1) (Score: 15%) For the first data point $\mathbf{x}_1 = [0.1, -0.1]^T$ from the training set, what is the output value of the neural network? Show your intermediate calculations.
2) (Score: 15%) Assuming the output of the network is $y_1 = 0.38$ given the data point $\mathbf{x}_1$. What is the cross entropy error function value $E(t_1, y_1)$, if the corresponding target is $t_1 = 1$.
3) (Score: 40%) Given the data point $\mathbf{x}_1$ and the target t_1, use backpropagation to calculate the following partial derivatives of the error: $\frac{\partial E(t_1,y_1)}{\partial w_{1,1}^{(2)}}$, $\frac{\partial E(t_1,y_1)}{\partial w_{2,4}^{(1)}}$ Show your intermediate calculations.
4) (Score: 15%) Briefly explain how you will use $\frac{\partial E(t_1,y_1)}{\partial w_{2,4}^{(1)}}$ to update the weight $w_{2,4}$ of the neural network for the 2nd iteration.
5) (Score: 15%) Briefly explain how this network output could be used for a binary classification task, where we must assign a test input x_n to either class $\mathcal{C}_0$ ($t_n = 0$) or class $\mathcal{C}_1$ ($t_n = 1$). Specifically, if the network output for a test input is $y = 0.38$, which class will you assign to that input and why?